



UrbanCool AI

URBANCool AI

Smart City Decision Support Platform for
Urban Heat Island (UHI) Mitigation



ANALYZE
Heat Risk
Hotspots



SIMULATE
What-If
Scenarios



EXPLAIN
AI Insights
with SHAP



OPTIMIZE
ROI Driven
Solutions



CONSULT
AI Planner
Assistant



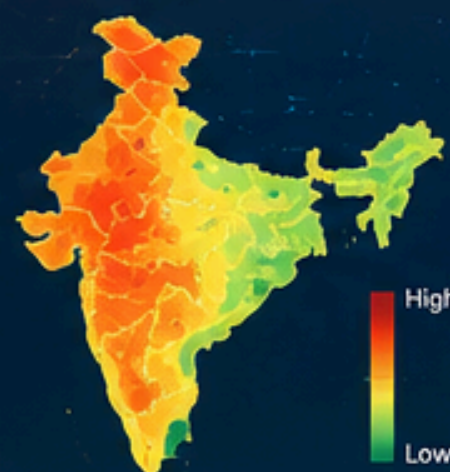
COOLER CITIES. SUSTAINABLE FUTURE.

Data Driven. AI Powered. Impact Focused.

AI FOR
COOLER
TOMORROW



India Heat Map



Digital Twin Simulation

Before



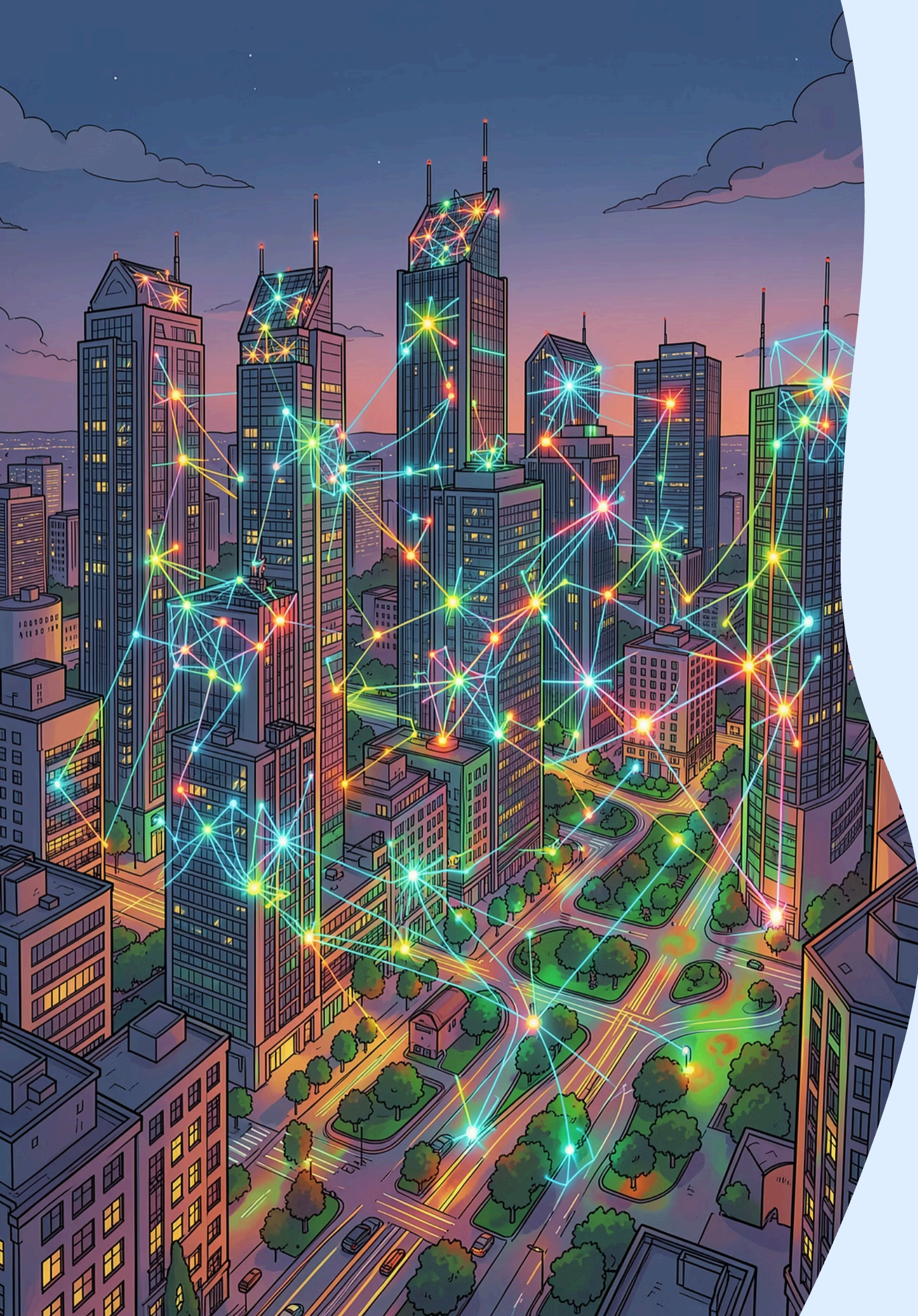
After



Heat Risk Index



High Risk



URBAN COOL AI

AI-Powered Decision Support for Urban Heat Island Mitigation

GDG WOW FEST - 2026

Predict • Simulate • Optimize • Cool Cities

Urban Heat Island (UHI) Problem



Rapid Urbanization

Cities expand faster than green infrastructure, creating heat-trapping concrete landscapes.

Declining Green Cover

Loss of trees and vegetation directly raises Land Surface Temperature (LST).

Reactive Planning

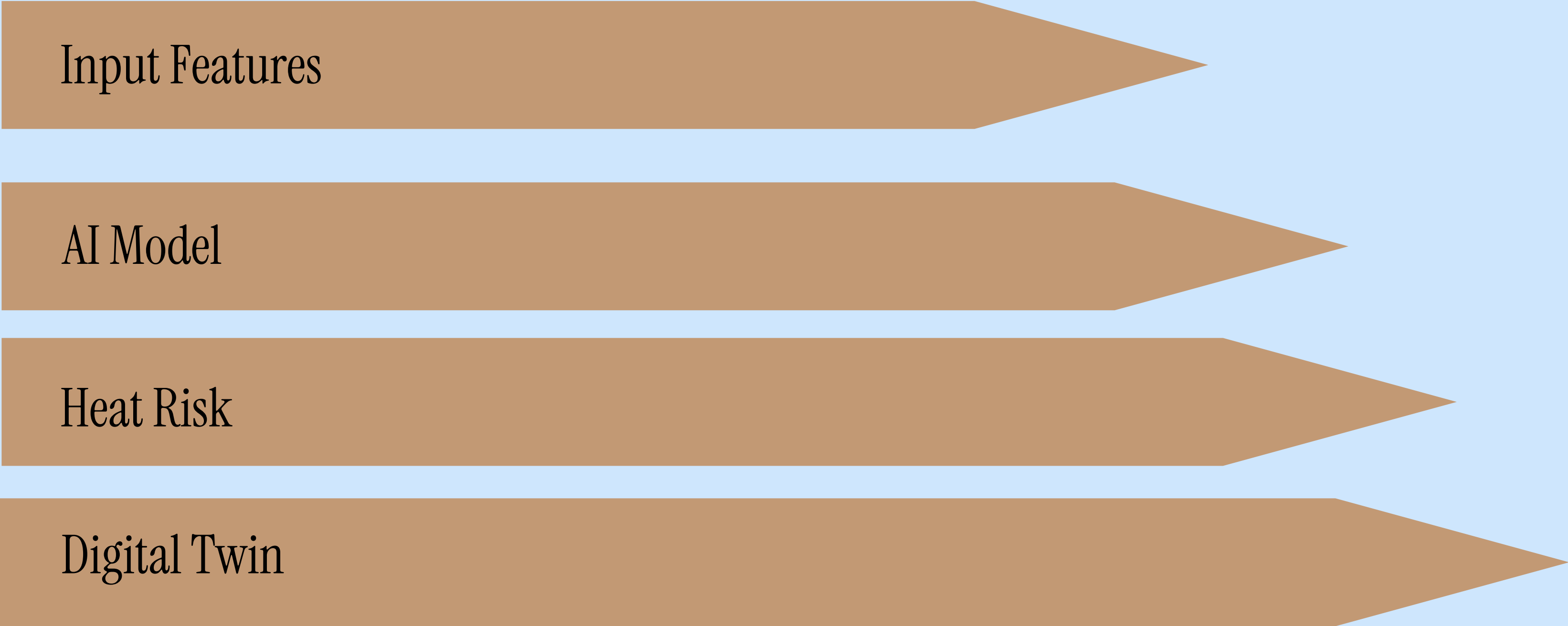
Traditional methods respond to heat events rather than predicting and preventing them.

Energy & Health Costs

Higher temperatures spike electricity demand and increase heat-related health risks.

Proposed Solution: UrbanCool AI

UrbanCool AI enables city planners to move from reactive to predictive, data-driven decision-making — simulating urban changes before they happen.



Input Features

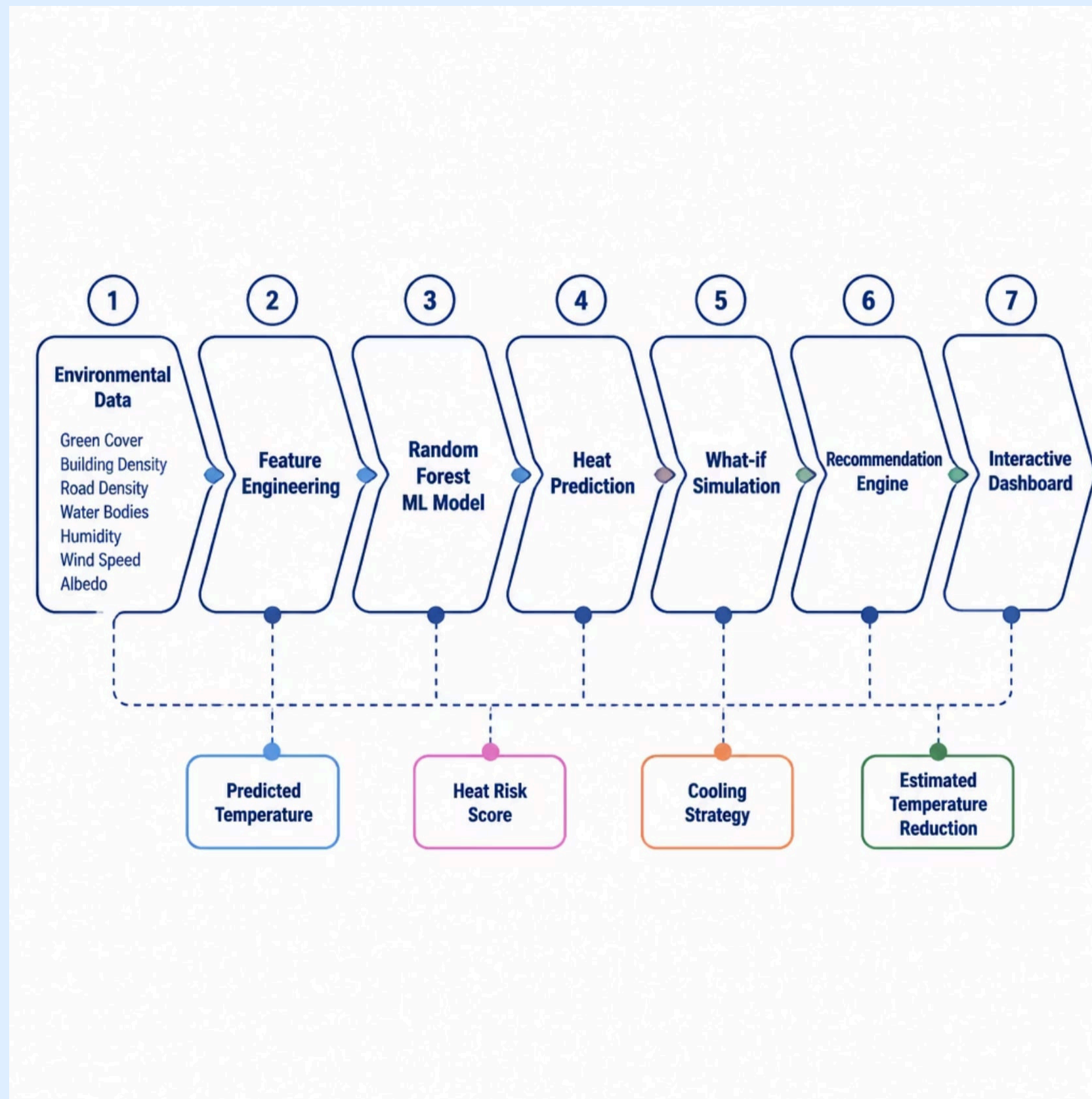
AI Model

Heat Risk

Digital Twin

Each stage transforms raw environmental data into actionable, explainable cooling strategies — all within a single integrated platform.

How UrbanCool AI Works



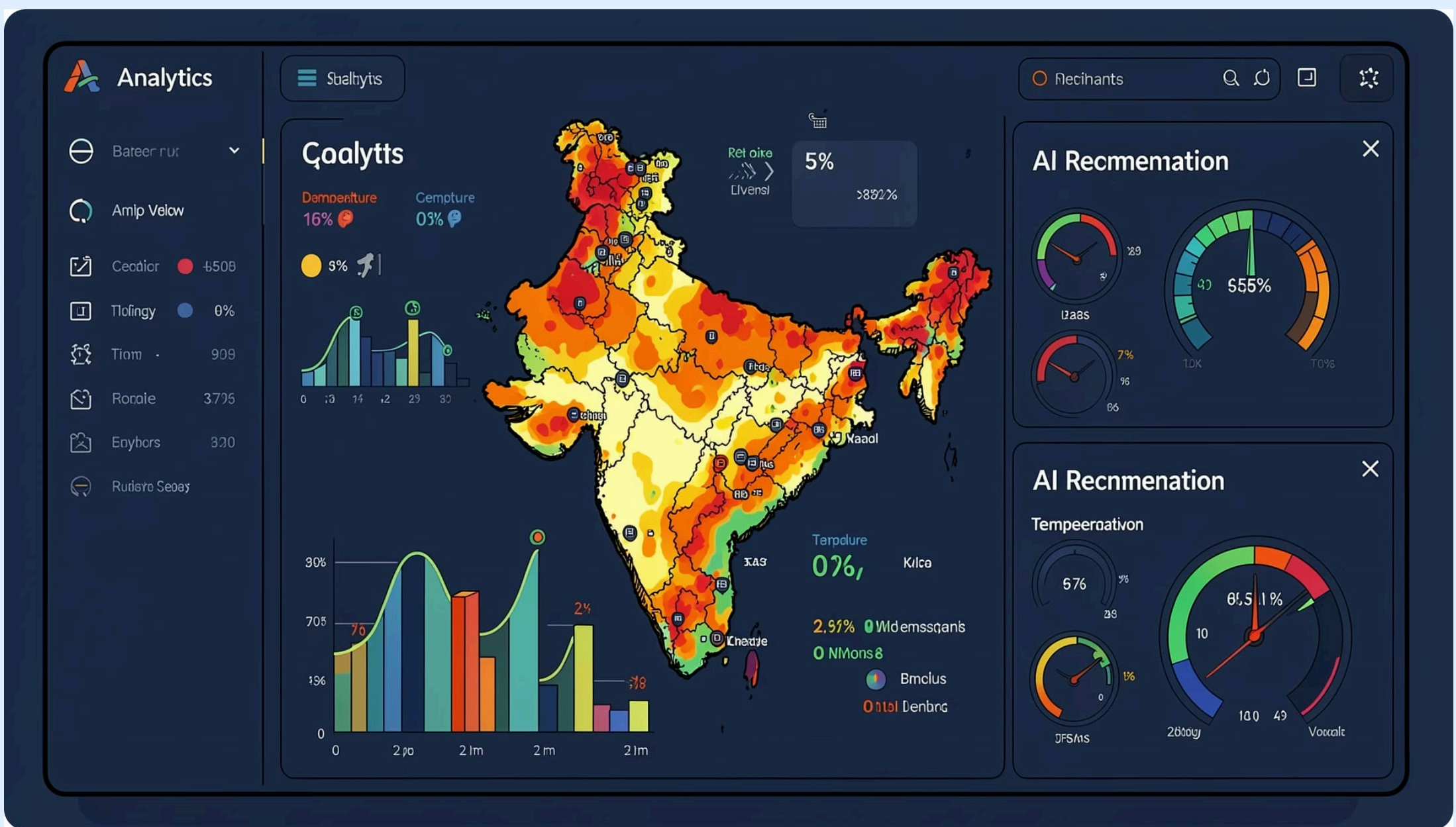
Input Features

- Green Cover & NDVI
- Building & Road Density
- Water Bodies & Albedo
- Humidity & Wind Speed

Outputs

- Predicted Land Surface Temperature
- Heat Risk Score
- Recommended Cooling Strategy
- Estimated Temperature Reduction

Interactive Smart City Dashboard



India Heat Map

City-wise LST analysis with ward-level granularity.



Digital Twin Simulation

What-if scenario planning before physical implementation.



SHAP Explainable AI

Transparent, interpretable model outputs for stakeholders.



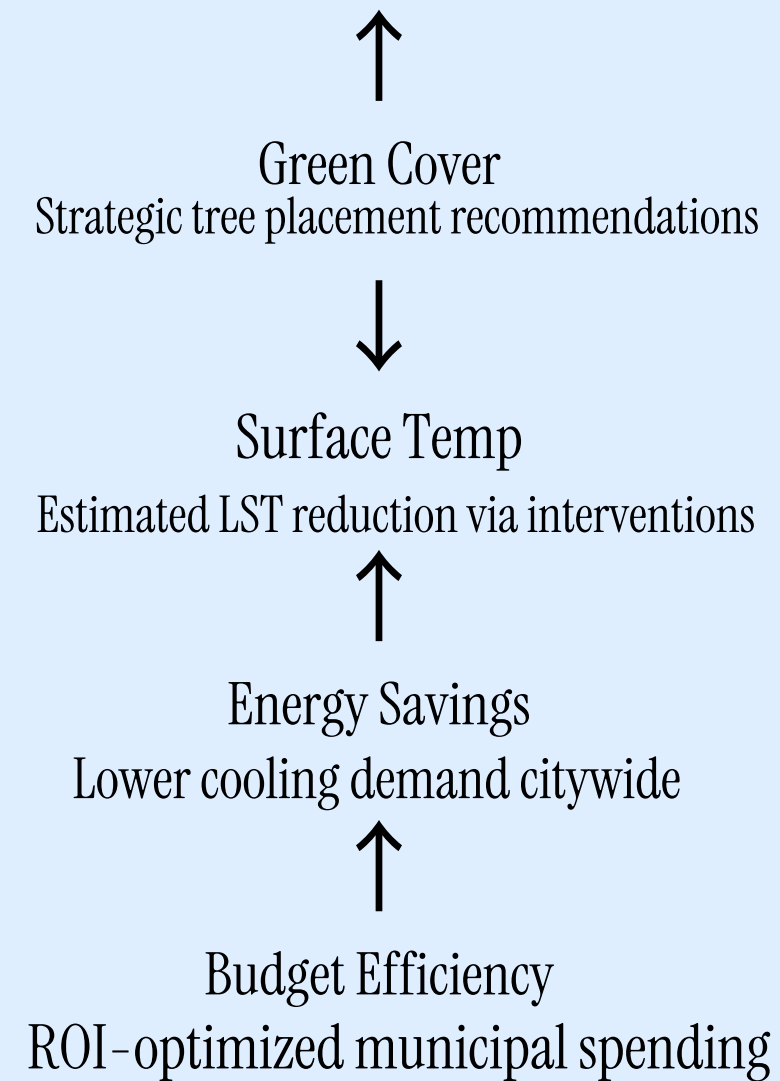
ROI-Based Recommendations

Budget-optimized cooling strategies with cost-benefit analysis.



Real-time simulation enables planners to evaluate cooling strategies before implementation.

Results & Benefits

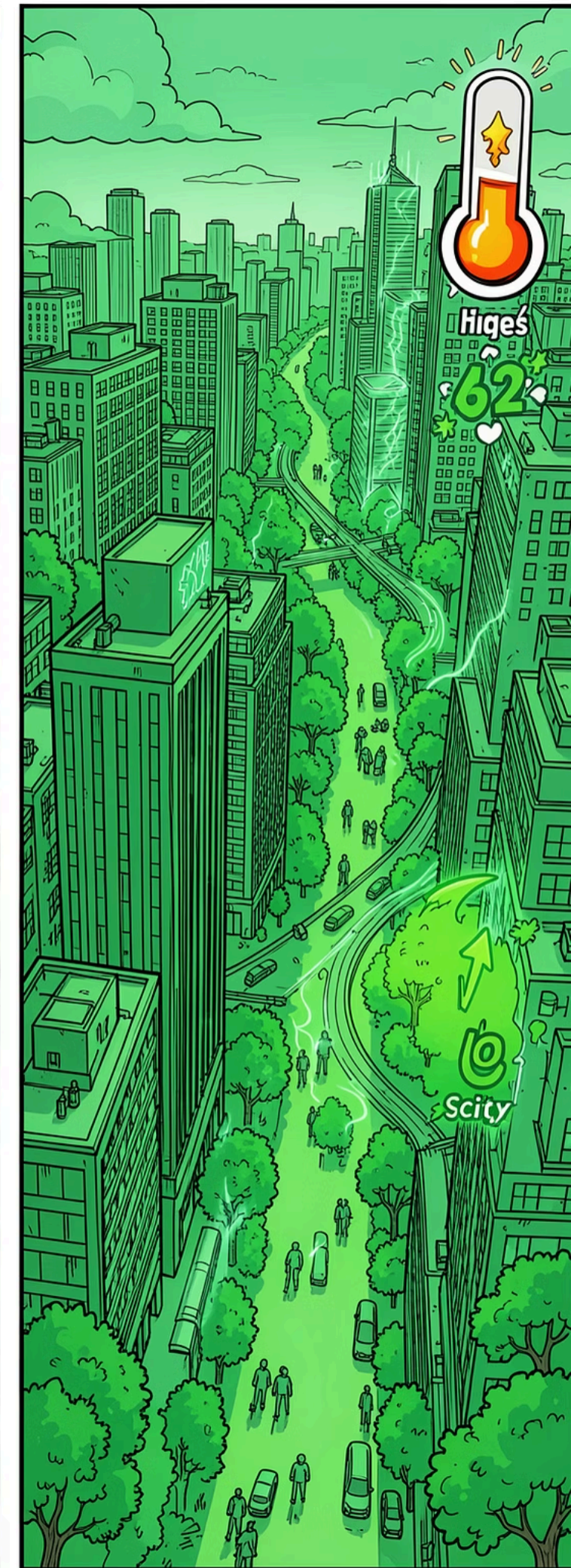
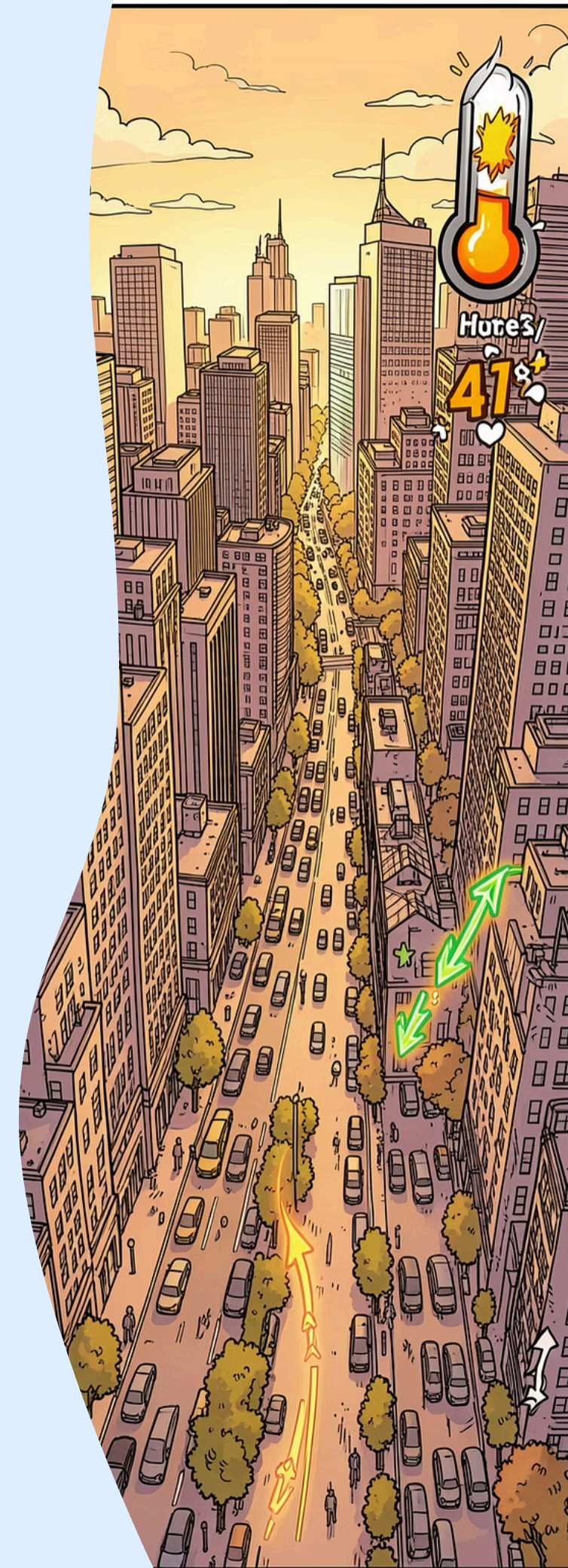


Technical Benefits

- Real-time prediction engine
- Explainable AI (SHAP)
- Fast what-if simulation
- Scalable cloud architecture

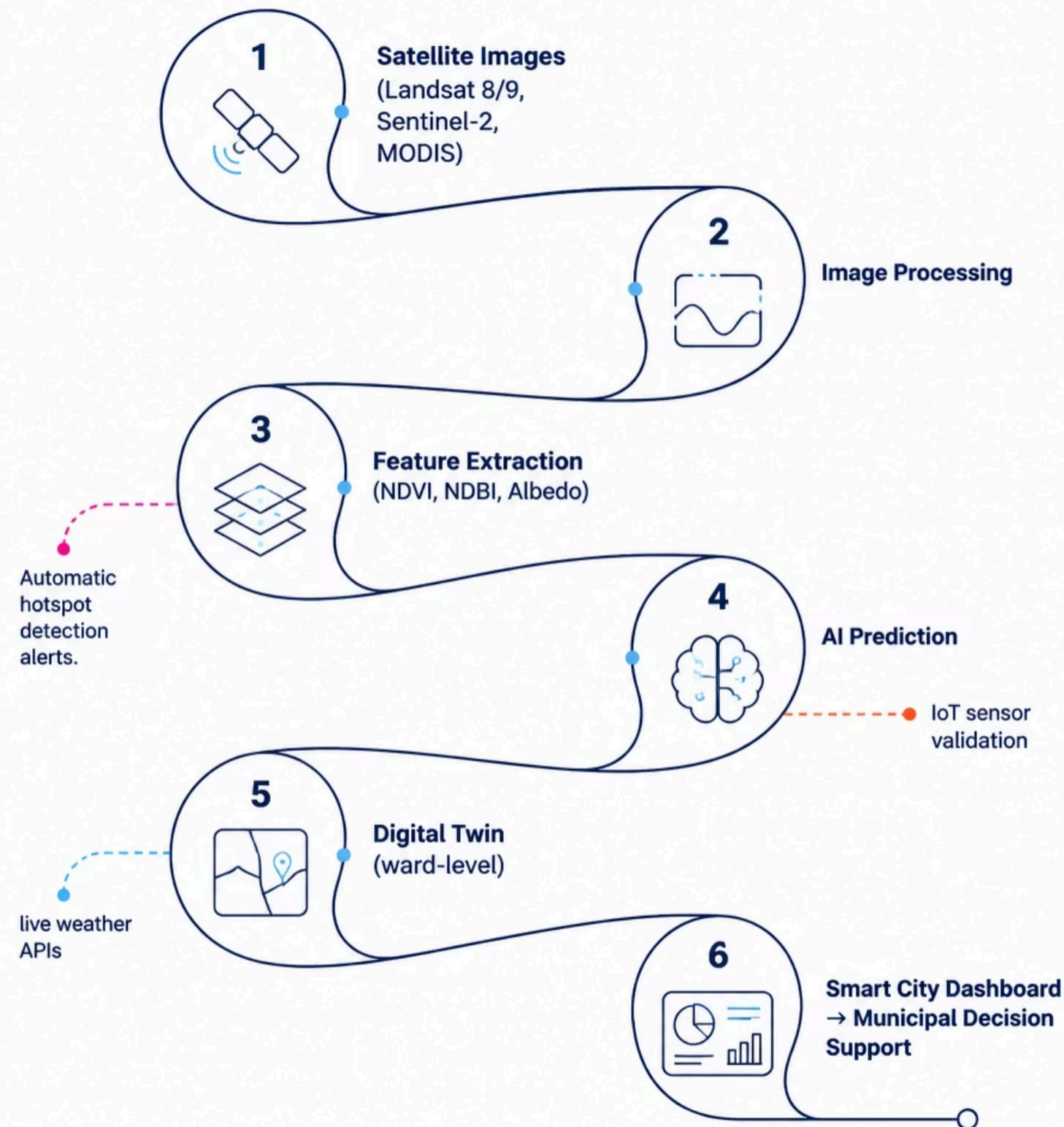
Environmental Benefits

- Reduced UHI effect
- Expanded green infrastructure
- Lower electricity consumption
- Improved public health outcomes



Future Enhancements: Satellite-Powered Smart Urban Planning

The current MVP uses synthetic/tabular environmental data. The next version integrates real satellite imagery for real-world municipal deployment.



Key Upgrades

- Live satellite imagery — Landsat 8/9, Sentinel-2, MODIS for accurate LST
- NDVI, NDBI & Albedo extracted from satellite data
- Live weather APIs for continuous environmental monitoring
- Ward-level Digital Twin of entire cities
- Automatic hotspot detection using thermal imagery
- AI heat alerts for municipal authorities
- IoT sensor integration for ground truth validation
- Nationwide Smart City Decision Support Platform

"From Synthetic Data Today to Real-Time Satellite Intelligence Tomorrow."